

JOSHUA LU

Tustin, CA 92782

☎ 657-565-2531

✉ joshualu82@gmail.com

🌐 www.linkedin.com/in/joshualu0201

🐙 github.com/GOGOJO

Education

University of California, Los Angeles

Expected Dec 2027

Bachelor of Science in Mathematics of Computation

Los Angeles, CA

Relevant Coursework

- Data Structures
- Linear Algebra
- Artificial Intelligence
- Computer Math
- Probability/Statistics
- Computer Architecture
- Front-End Development
- Software Development

Experience

Coding Minds

November 2023 – June 2025

Computer Science tutor

Irvine, CA

- Provide one-on-one and small group instruction in programming, algorithms, and data structures to students of varying skill levels, resulting in a **95% student retention rate** through engaging, comprehensive teaching methods.
- Designed individualized lesson plans to meet students' learning objectives, covering key computer science concepts such as **control flow, object-oriented programming, data structures and algorithms**.
- Guided students through hands-on projects including interactive games, automation scripts, and algorithm visualizations to reinforce understanding and practical application.

Whealthy

June 2023 – December 2023

Restaurant Manager

Irvine, CA

- Supervised daily operations of a full-service restaurant, including scheduling, inventory, and team supervision
- Managed a 12-member team with shift scheduling software, **improving efficiency by 20%**
- Implemented inventory tracking system that **reduced waste by 15%**

Projects

SpendShift | React, JavaScript, SQLite, FastAPI, HTML, CSS

- Developed a full-stack personal finance app using **React** for the frontend and **FastAPI + SQLite** for the backend, enabling users to register/log in, track transactions, and set savings goals.
- Implemented JWT-based authentication and protected routes to secure user data, ensuring a robust sign-up/login flow and restricted access to dashboard and goal features.
- Designed and built RESTful API endpoints for CRUD operations (transactions and goals) using SQLAlchemy in FastAPI, enabling users to create, view, update, and delete their data seamlessly.

Discord Bot | Python, discord.py, FFmpeg, yt-dlp

- Built a Discord music bot that streams audio directly into voice channels using the Discord API.
- Integrated yt-dlp and FFmpeg to download and stream music from YouTube in real-time.
- Implemented commands such as !play, !pause, !skip, and !queue using discord.py event handlers.
- Deployed Python-based Discord music bot supporting 30+ concurrent users with real-time audio streaming

Syllabus to Calendar | Python, NLP, Google Calendar API, Render

- Developed a tool that parses course syllabi and automatically extracts assignment, midterm, and final exam dates from unstructured text.
- Implemented text parsing and date extraction using regular expressions and natural language processing techniques to identify deadlines across multiple course formats.
- Integrated the Google Calendar API to automatically generate and insert events, reducing manual schedule management for students.
- Designed a scalable pipeline to process multiple syllabi and convert academic schedules into structured calendar data.

LeetTrack | Next.js, TypeScript, Prisma, PostgreSQL, NextAuth, Tailwind, Recharts

- Built a full-stack LeetCode analytics dashboard that syncs public profile data via the LeetCode GraphQL API, with OAuth sign-in and persisted user profiles in PostgreSQL using Prisma.
- Implemented a weighted interview-readiness score and topic-level analytics (arrays, graphs, DP, etc.) to surface weak areas from solve counts, acceptance rate, and difficulty mix.
- Designed interactive dashboards with Recharts (submission heatmap, progress over time, difficulty distribution) and daily goal tracking with server-side sync caching.

Technical Skills

Languages: Python, JavaScript, Java, Assembly, C, C++, C#, HTML/CSS, Go, Agile, R, TypeScript, Next.js

Developer Tools: VS Code, Linux, Bash, Unity, Git, GitHub, Google Cloud

Frameworks: React, Numpy, Pandas, Matplotlib, Scikit-learn, Flask, Django, Pygame, Node.js, FFmpeg, Angular